

Rep Smarter, Not Harder: AI Hypertrophy Coaching with Wearable Sensors and Edge Neural Networks

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Abstract—abstract

I. INTRODUCTION

A. *Background*

Exercise science background

B. *Related Work*

Literature review

C. *Contributions*

What we did new

II. MATERIALS AND METHODS

A. *Data Collection*

1) *Participants:*

2) *Equipment:* data collection tool (Wearable sensors)

3) *Procedure:* instructions, data annotation

B. *Data Processing*

data cleaning, windowing, splitting, augmentation

C. *Segmentation Model Design*

D. *Classification Model Design*

E. *Model Training*

F. *Model Evaluation*

G. *Edge Deployment*

describe steps taken to deploy to edge

III. RESULTS

Put any figures and tables here

A. *Segmentation Model Performance*

B. *Classification Model Performance*

C. *Edge Deployment Performance*

IV. DISCUSSION

V. REFERENCES